ECON120B: Final Exam Review

Content

- OLS and estimators, and Causal treatment effect (before midterm)
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- Threats to internal validity
  - Omitted variable bias
  - Errors-in-variables
  - Sample selection
  - Simultaneous causality
- Threats to external validity

OLS and estimators (before midterm)

Check if you know the following

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Know the following:

- 3 OLS Assumptions (Key Concept 4.3)
- Formulas for $\hat{\beta}_1$ and $\hat{\beta}_0$ (Key Concept 4.2)
- Regression Output : ESS, TSS, SER, RMSE, R^2, $\hat{Y}_i$, $\hat{u}_i$, SE($\hat{\beta}_1$)
- Hypothesis testing on β_1 (two-sided and one-sided)
- Heteroskedasticity and Homoskedasticity
- What Assumptions are required for consistency?
- What Assumptions are required for Gauss-Markov Theorem?

Causal treatment effect (before midterm)

Check if you know the following

$$ATE = E[Y_i^1 - Y_i^0]$$
$$ATT = E[Y_i^1 - Y_i^0 | D_i = 1]$$

Know the following:

- How to compute ATE and ATT
- How to compute observed difference and selection bias
- Also could think of sign of selection bias
  - For example, aspirin: Individuals who chose to take aspirin would have higher headache than others who didn't choose aspirin if both of two groups didn't take aspirin
  - Selection bias : $E[Y_i^0|D_i = 1] - E[Y_i^0|D_i = 0]$ is positive by logic above (Here outcome is level of headache)

Decomposing Observed Difference into ATE and Selection Bias

$$\underbrace{E[Y_i|D_i=1]-E[Y_i|D_i=0]}_{\text{Observed difference}} = E[Y_i^1|D_i=1]-E[Y_i^0|D_i=0]$$
$$= \underbrace{E[Y_i^1|D_i=1]-E[Y_i^0|D_i=1]}_{\text{ATT}}$$
$$+ \underbrace{E[Y_i^0|D_i=1]-E[Y_i^0|D_i=0]}_{\text{Selection bias } (=0)}$$

RCT, or randomized experiments

- Random: Subjects *randomly* assigned to treatment and control groups
- Controlled: There is a *control* group which doesn't receive treatment
- Trial: Treatment is assigned with subjects having no choice

In RCT, treatment, D_i is *independent* of Y_i^1 and Y_i^0. Hence,

$$E[Y_i^1|D_i = 1] = E[Y_i^1|D_i = 0] = E[Y_i^1]$$
$$E[Y_i^0|D_i = 1] = E[Y_i^0|D_i = 0] = E[Y_i^0]$$

In RCT, ATE and ATT are equal to observed difference

$$\underbrace{E[Y_i|D_i=1]-E[Y_i|D_i=0]}_{\text{Observed difference}} = E[Y_i^1|D_i=1]-E[Y_i^0|D_i=0]$$
$$= \underbrace{E[Y_i^1]-E[Y_i^0]}_{\text{ATE}}$$

Multiple Regression Model

How to interpret coefficients

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

How to interpret coefficients:

- β_0 is population intercept, expected value of Y_i when all Xs are 0
- Other β's - change in Y_i for a 1 unit change in the associated X_i (holding all other X's constant
  - In math terms - $\beta_1 = \Delta Y / \Delta X_1$, **holding** $X_2, X_3, \ldots, X_k$ **constant**
- If X is a dummy variable, coefficient (β) is the difference in expected Y between the category ($X = 1$) and the omitted group ($X = 0$)

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Least Square Assumptions for Multiple Regression:

1. $E(u|X_1 = x_1, \ldots, X_k = x_k) = 0$
2. $(X_{1i}, \ldots, X_{ki}, Y_i)$, $i = 1, 2, \ldots, n$, are i.i.d.
3. Large outliers are rare
   - Or, $E(X_{1i}^4) < \infty, \ldots, E(X_{ki}^4) < \infty, E(Y_i^4) < \infty$.
4. There is no perfect multicollinearity ($\leftarrow$ New!)

$X_1, X_2, \ldots, X_k$ are *perfectly multicollinear* if we can find numbers $a_1, a_2, \ldots, a_k$ such that

$$X_1 = a_1 + a_2 X_2 + \cdots + a_k X_k$$

Example: Age Ranges

- $Young = 1$ if $Age < 25$, and $Young = 0$ otherwise
- $MiddleAge = 1$ if $25 \leq Age < 55$, and $MiddleAge = 0$ otherwise
- $Old = 1$ if $55 < Age$, and $Old = 0$ otherwise

Young, MiddleAge, Old are mutually exclusive and exhaustive, which means you must be one of three groups. Hence we have the following relation:

$$Young = 1 - 1 \times MiddleAge - 1 \times Old$$

Check if this works by plugging in possible values for *MiddleAge* and *Old*

What is wrong with the following regression?

$$Y_i = \beta_0 + \beta_1 Young_i + \beta_2 MiddleAge_i + \beta_3 Old_i + u_i$$

Due to perfect multicollinearity of *Young*, *MiddleAge*, *Old*, so violation of assumption (4)

Solution: Exclude one of the categories. For example,

$$Y_i = \beta_0 + \beta_1 Young_i + \beta_2 MiddleAge_i + u_i$$

- *Old* is the excluded category (reference group)
- β_1 is average of *Young* relative to excluded category *Old*
- β_2 is average of *MiddleAge* relative to excluded category *Old*
- β_0 is average of excluded category *Old*
- Interpretation and value of coefficients changes depending on which category you exclude
- What is the interpretation of β_1 if we exclude the interept instead?

Imperfect multicollinearity:

- Occurs when regressors are highly correlated
- Does *not* violate assumption (4) or imply OVB (OVB : explained later)
- Increases the standard errors

Multiple Regression Model, again:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Examples of Joint Null Hypothesis:

- $\beta_1 = 0, \beta_2 = 0, \ldots, \beta_k = 0 \rightarrow q = k$
- $\beta_1 = 0, \beta_2 = 0 \rightarrow q = 2$
- $\beta_1 = \beta_2 \rightarrow q = 1$

$q =$ number of restrictions (count equal signs)

Should know how to set up null and alternative hypotheses

- Example : $H_0 : \beta_1 = 0, \beta_2 = 0,\ H_1 :$ either $\beta_1 \neq 0$ or $\beta_2 \neq 0$ or both

Example: $birthweight_i = \beta_0 + \beta_1 Smoker_i + \beta_2 alcohol_i + \beta_3 nprevist_i + u$

```
. reg birthweight smoker alcohol nprevist, robust

Linear regression                          Number of obs     =      3,000
                                           F(3, 2996)        =      59.48
                                           Prob > F          =     0.0000
                                           R-squared         =     0.0729
                                           Root MSE          =     570.47
```

birthweight	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
smoker	-217.5801	26.10764	-8.33	0.000	-268.7708	-166.3894
alcohol	-30.49129	72.59671	-0.42	0.675	-172.8357	111.8531
nprevist	34.06991	3.608326	9.44	0.000	26.99487	41.14496
_cons	3051.249	43.71445	69.80	0.000	2965.535	3136.962

```
. test alcohol nprevist

 ( 1)  alcohol = 0
 ( 2)  nprevist = 0

       F(  2,  2996) =   44.71
```

- $H_0 : \beta_2 = 0, \beta_3 = 0$
- $H_1 : \beta_2 \neq 0$, or $\beta_3 \neq 0$, or both $(q = 2)$
- F-statistic is 44.71, larger than 5% critical value of $\chi^2_2/2$, 3.00.
- Reject the null that *alcohol* and *nprevist* don't have effect on *birthweight*.

Non-linear models

What if the relationship between X and Y is NOT linear?

- Then linear regression $Y_i = \beta_0 + \beta_1 + X_i + u_i$ leads to biased estimator of effect on Y of X

How to solve this problem? : use a nonlinear function in regression

- Log transforms of Y and/or X
- Polynomials in X: for example, quadratic formula is as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

- Interactions between regressors

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 (X_i \times Z_i) + u_i$$

Non-linear models

Interpret variables with log form

How to interpret change in variable in log form?

- $ln(X)$ increases by 0.01 $\approx$ X increases by 1%
- For example, if $ln(X)$ increases by 0.025, then X goes up by 2.5%

$$(\log)Y_i = \beta_0 + \beta_1(\log)X + u_i$$

	Y	**log Y**
Dummy X	for $X = 1$ compared to $X = 0 \implies Y \uparrow \beta_1\{\text{units}\}$	for $X = 1$ compared to $X = 0 \implies Y \uparrow 100\beta_1\%$
X	$X \uparrow 1\{\text{units}\} \implies Y \uparrow \beta_1\{\text{units}\}$	$X \uparrow 1\{\text{units}\} \implies Y \uparrow 100\beta_1\%$
log X	$X \uparrow 1\% \implies Y \uparrow \beta_1/100\{\text{units}\}$	$X \uparrow 1\% \implies Y \uparrow \beta_1\%$

Why do we multiply by 100? Because the coefficient β_1 is measured in shares. To get percentages we need to multiply the respective side by 100.

$$\text{wage}_i = \beta_0 + \beta_1 \text{education}_i + 1000\text{age}_i - 10\text{age}_i^2 + u_i$$

- How do we test for linearity in age? $H_0 : \beta_3 = 0$ vs. $H_1 : \beta_3 \neq 0$
- Interpretation: The marginal effect would depend on a point, e.g.: effect of age change from 30 to 31: $1000 \times (31 - 30) - 10 \times (31^2 - 30^2) = 400$
- Would a marginal effect decrease or increase with age?

Regression with interaction term of continuous variable and binary variable

$$\text{Income}_i = \beta_0 + \beta_1 \text{male}_i + \beta_2 \text{age\_of\_child}_i + \beta_3 \text{male}_i \times \text{age\_of\_child}_i + u_i$$

Consider two cases:

- $male = 0$: $\widehat{Income} = \hat{\beta}_0 + \hat{\beta}_2 age\_of\_child$
- $male = 1$:

$$\widehat{Income} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 age\_of\_child + \hat{\beta}_3 age\_of\_child = \underbrace{(\hat{\beta}_0 + \hat{\beta}_1)}_{\text{intercept}} + \underbrace{(\hat{\beta}_2 + \hat{\beta}_3)}_{\text{slope}} age\_of\_child$$

Interaction term between variable X_1 and X_2 is interpreted as how X_2 changes the effect of X_1.

$$\text{Income}_i = \beta_0 + \beta_1 \text{male}_i + \beta_2 \text{age\_of\_child}_i + \beta_3 \text{male}_i \times \text{age\_of\_child}_i + u_i$$

- **Interpretation of β_1:** A gap in income between men and women at child birth is β_1 dollars.
- **Interpretation of β_3:** Each additional year of child increases a gap in income by β_3 dollars.

$$\text{Income}_i = \beta_0 + \beta_1 \text{male}_i + \beta_2 \text{age\_of\_child}_i + \beta_3 \text{male}_i \times \text{age\_of\_child}_i + u_i$$

Can test various hypotheses:

- Test that effect of *age_of_child* is the same for men and women?
    - which means slope is same across *male*, so $H_0 : \beta_3 = 0$
- Test that the regression lines have the same intercept between men and women?
    - $H_0 : \beta_1 = 0$
- Test that the regression lines are the same?
    - which means slope and intercept are same across *male*, so $H_0 : \beta_1 = 0, \beta_3 = 0$
    - Need joint hypotheses testing so use an F-statistic (Here, $q = 2$.)

Another example:

$$\text{Income}_i = \beta_0 + \beta_1 \text{firm\_size}_i + \beta_2 \text{years\_of\_education}_i + \\ \beta_3 \text{firm\_size}_i \times \text{years\_of\_education}_i + u_i$$

- **Interpretation of β_1:** Each additional employee increases the income of other employees by β_1 dollars.
- **Interpretation of β_3:** Each additional year of education increases the effect of firm size on income by β_3 dollars per additional employee.

Threats to Internal Validity

- Four sources of threats to internal validity
  - **Omitted variable bias**
  - Errors-in-variables
  - Sample selection
  - Simultaneous causality

Threats to Internal Validity

Two conditions for OVB to occur:

1. The omitted variable is correlated with the included regressor
2. The omitted variable is a determinant of the dependent variable

In mathematical form, consider a regression with omitted variable Z,

$$Y_i = \beta_0 + \beta_1 X_i + u_i = \beta_0 + \beta_1 X_i + (\gamma Z_i + v_i)$$

Then two conditions are as follows,

1. $Cov(X, Z) \neq 0$
2. $\gamma \neq 0$

Which OLS assumption is violated if we have OVB?
Assumption 1: $E[u_i | X_i = x_i] = 0$

The omitted variable bias formula:

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \gamma \frac{Cov(X, Z)}{Var(X)}$$

Hence, sign of OVB is determined by $Cov(X, Z)$ and γ.

Example: Effect of being in preschool on earning in adulthood

$$Earning_i = \beta_0 + \beta_1 Preschool_i + u_i$$

Here *Preschool* is 1 if an individual i was in preschool in childhood, and 0 otherwise. Does *Preschool* suffer by omitted variable bias problem?

Omitted variable : Parents' Income

1. $Cov(Preschool, Parents'\ Income) > 0$ because Preschool is expensive
2. $\gamma > 0$, Parents' income might have a positive impact on earning

Sign of OVB : Sign(γ) $\times$ Sign($Cov(X, Z)$) $= (+) \times (+) = (+)$

- This means $\hat{\beta}_1 > \beta_1$
- $\hat{\beta}_1$ will overstate the change in earning associated with going to preschool

How to solve omitted variable bias problem?

- Randomized Controlled Experiment (usually infeasible)
    - In the previous example, make treatment (being in a preschool) being randomly assigned to each individual, so that there is no correlation between being in a preschool and parents' income
    - This way could make $Cov(X, Z) = 0$, even though γ is still nonzero
- **Include the omitted variable in the regression**

Threats to Internal Validity

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Threats to Internal Validity

- Four sources of threats to internal validity
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How can it arise?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

What is effect on estimators if ...

- X is measured with error?
    - $\hat{\beta}_1$ is biased and inconsistent! Specifically biased *towards zero*.
    - Intuition: In extreme case, X is all noise $\rightarrow$ no effect of X on Y, which means $\beta_1 = 0$
- Y is measured with error?
    - $\hat{\beta}_1$ is still unbiased and consistent (assumed error uncorrelated with X)
- Check Exercise 9.2 in the textbook

How to solve this problem? : Get a better data if possible, develop a specific model of errors, or instrumental variable regression

Threats to Internal Validity

- Four sources of threats to internal validity
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  - **Sample selection**
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Threats to Internal Validity

Sample selection : When this is a problem

Sample selection bias arises when a selection process:

1. influences the availability of data and
2. that process is related to the dependent variable

Example: Effect of a 6 month Diet program. 1000 people, half in a diet program and half not. Interested in the effect on Weight Loss of the Diet program, so you set up the regression as follows.

$$WeightLoss_i = \beta_0 + \beta_1 Diet_i + u_i$$

You sample only people who *finish* Diet program, as well as those not in the diet program: What's the problem of this sampling process?

You sample only people who *finish* Diet program, as well as those not in the diet program: What's the problem of this sampling process?

- A person might prematurely leave the Diet if it isn't working, so being in the program means the diet worked well for you
- Lost the people for whom the diet didn't work (availability of data)
- Have people who succeed to lose weight by program only (related to dependent variable)
- Hence $\hat{\beta}_1$ will overstate the effect of the program

How to solve this problem? : Collect the sample in a different way, RCT, or develop a model of sample selection and estimate this model

Threats to Internal Validity

- Four sources of threats to internal validity
  - Omitted variable bias
  - Errors-in-variables
  - Sample selection
  - **Simultaneous causality**

I want to know X's causal effect on Y, *but* Y also affects X

Example: Effect of smoking (X) on mental health (Y)

- Smoking (X) would affect one's mental health (Y)
- If a person is stressed (Y), however, likely to smoke more (X)
- In this case, $\hat{\beta}$ is biased and inconsistent

How to solve this problem? : RCT, or instrumental variable regression

Threats to External Validity

Two sources of threats

Can we apply the regression results obtained from one context to another?

Difference in population

- Is your sample close enough to another group of population? or similar to general population?
- i.e. Animal and human, people in different states, college graduates and general people, etc.

Difference in Setting

- Is there any difference in settings? i.e. institutional environment, laws, or physical environment
- i.e. public or religious universities, price change in tobacco, etc.